

$$\begin{cases} 2x_1 + x_2 = 7 \\ x_1 + x_2 - 3x_3 = -10 \\ 6x_2 - 2x_3 + x_4 = 7 \\ 2x_3 - 3x_4 = 13 \end{cases}$$

Quantitative Aptitude

Module I: Algebra

Indices, Surds, Logarithms – Introduction and Application, Types of Equations – Linear and Quadratic, Roots of Quadratic Equations, Methods to solve Quadratic Equations, Simultaneous Equations - Methods to solve (two or three unknowns), Inequalities with graphs

Module II: Data Arrangement

Arithmetic and Geometric Progressions – Definition, finding term, Sum till term, arithmetic and geometric means, Applications in real life, Permutation and Combination – Introduction and Applications

Module III: Relations and functions, Limit and Continuity

Sets – Definition, Forms of writing sets, Types of sets, Relations and Functions – Definition, Types of functions, Limit of a function – Definition and Methods to solve, Continuous and Discontinuous functions

Module IV: Data Analysis

Data – Definition, Frequency distribution, Graphical representation, Histogram, Frequency polygon, Frequency curve and Ogive, Measures of central tendency, Measures of Dispersion, Skewness and Kurtosis

Module V: Forecasting Techniques

Correlation Analysis – Introduction, methods to find correlation coefficient,
Regression Analysis – Linear regression, Method of least square